Homework 6

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We used the tidyverse (H. Wickham *et al.* [1]) , knitr (Y. Xie [2]) , and here (K. Müller [3]) packages as well as some additional information on the star data (E. Hanushek [4]).

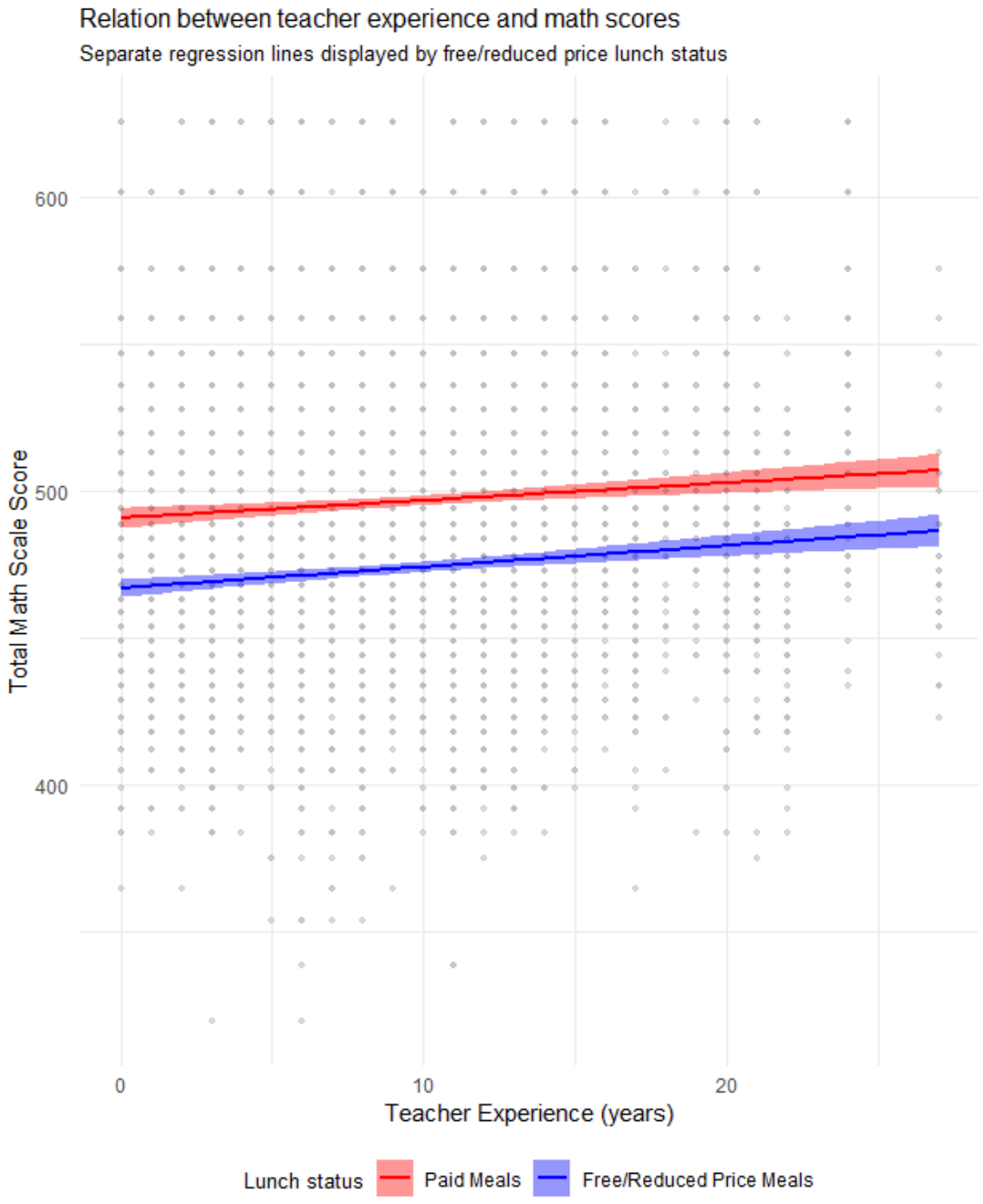
References

Sex	FRL	Math Mean	Math SD	Reading Mean	Reading SD
boy	no	492.85	46.34	441.46	32.32
boy	yes	469.87	46.09	425.38	26.63
girl	no	501.21	45.96	448.54	34.52
girl	yes	477.51	46.30	430.80	27.42

#The summary statistics table above shows that students who were not eligible for free or reduced lunch had higher average math and reading scores than students who were eligible for free or reduced lunch. Girls had higher average scores than boys in both subjects across free and reduced lunch categories. The standard deviations shows variability in both math and reading scores.

Relationship Between Teacher Experience and Math Scores

The plot below shows the relationship between teacher experience and students' mathematics scores, with separate regression lines for students receiving free/reduced-price meals and those receiving paid meals.



The plot shows a slight positive relationship between teacher experience and students' mathematics scores. Students whose teachers have more years of experience tend to have slightly higher math scores. Students who pay for meals generally have higher math scores than those receiving free or reduced-price meals, regardless of teacher experience.

Bibliography

[1] H. Wickham *et al.*, "Welcome to the tidyverse," *J. Open Source Softw.*, vol. 4, no. 43, p. 1686, Nov. 2019, doi: 10.21105/joss.01686.

[2] Y. Xie, "knitr: A General-Purpose Package for Dynamic Report Generation in R." 2026. [Online]. Available: https://yihui.org/knitr/

[3] K. Müller, "here: A Simpler Way to Find Your Files." 2025. [Online]. Available: https://here.r-lib.org/

[4] E. Hanushek, "Some Findings From an Independent Investigation of the Tennessee STAR Experiment and From Other Investigations of Class Size Effects," vol. 21, 1999, doi: 10.3102/01623737021002143.